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Heat dissipation structure and electronic device

Abstract

Provided is a heat dissipation structure, including a conductive member disposed on and thermally coupled to a heat source, a heat pipe including evaporating and condensing parts, a fan disposed in correspondence to the condensing part, and a heat storage component disposed on a circuit board. The evaporating part is disposed on and thermally coupled to the conductive member. The heat source is located between the conductive member and the circuit board. The conductive member is located between the heat pipe and the heat source. The circuit board is located between the heat source and the heat storage component, and is thermally coupled to the heat source. The heat storage component is thermally coupled to the circuit board and is filled with a working medium absorbing heat conducted from the heat source to the circuit board by latent heat of absorption during phase change. An electronic device is also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 110121330, filed on Jun. 11, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a heat dissipation structure and an electronic device, and more particularly to a heat dissipation structure and an electronic device using the heat dissipation

structure.

Description of Related Art

(3) With continuous improvements in computing performance of electronic devices, related manufacturers have been focusing on how to rapidly extract heat generated during operation of central processing units or graphics processing units inside the electronic devices.

(4) Generally speaking, the central processing unit or the graphics processing unit is connected to the circuit board, and the heat is extracted out by a remote heat exchanger (RHE), which transfers the heat sequentially through a conductive member thermally coupled to the surface of the central processing unit or the graphics processing unit and a heat pipe connected to the conductive member to a heat dissipation fin connected to the heat pipe, and finally dissipates the heat out of the system by heat exchange with the outside through forced convection provided by a fan.

(5) However, when the electronic device generates high-wattage transient power, relatively high thermal energy is generated instantly, and the existing RHE is unable to provide effective heat dissipation to the instantly generated relatively high thermal energy due to its heat dissipation rate.

SUMMARY

(6) The disclosure provides a heat dissipation structure, which helps enhance instant heat dissipation performance.

(7) The disclosure provides an electronic appliance, which has excellent instant heat dissipation performance.

(8) The disclosure proposes a heat dissipation structure, which includes a conductive member, at least one heat pipe, at least one fan, and at least one heat storage component. The conductive member is disposed on at least one heat source and is thermally coupled to the at least one heat source. The at least one heat pipe includes an evaporating part and a condensing part. The evaporation part of the at least one heat pipe is disposed on the conductive member and is thermally coupled to the conductive member. The at least one fan is disposed in correspondence to the condensing part of the at least one heat pipe. The at least one heat storage component is disposed on the circuit board. The at least one heat source is located between the conductive member and the circuit board. The conductive member is located between the at least one heat pipe and the at least one heat source. The circuit board is located between the at least one heat source and the at least one heat storage component. The circuit board is thermally coupled to the at least one heat source, and the at least one heat storage component is thermally coupled to the circuit board. The at least one heat storage component is filled with a working medium, and the working medium absorbs the heat conducted from the at least one heat source to the circuit board by latent heat of absorption during phase change.

(9) The disclosure provides an electronic device, which includes a case, a circuit board, at least one heat source, and a heat dissipation structure. The circuit board is disposed in the case. The at least one heat source is disposed on the circuit board. The heat dissipation structure includes a conductive member, at least one heat pipe, at least one fan, and at least one heat storage component. The conductive member is disposed on the at least one heat source. The conductive member is thermally coupled to the at least one heat source, and the at least one heat source is located between the conductive member and the circuit board. The at least one heat pipe includes an evaporating part and a condensing part. The evaporation part of the at least one heat pipe is disposed on the conductive member and is thermally coupled to the conductive member. The conductive member is located between the at least one heat pipe and the at least one heat source. The at least one fan is disposed in correspondence to the condensing part of the at least one heat pipe. The at least one heat storage component is disposed on the circuit board and is thermally coupled to the circuit board. The circuit board is located between the at least one heat source and the at least one heat storage component. The circuit board is thermally coupled to the at least one heat source, and the at least one heat storage component is thermally coupled to the circuit board. The at least one heat storage component is filled with a working medium, and the working medium

absorbs heat conducted from the at least one heat source to the circuit board by latent heat of absorption during phase change.

(10) Based on the above, in the heat dissipation structure and the electronic device of the disclosure, the circuit board is thermally coupled to the heat source, the heat storage component is further thermally coupled to the circuit board, and the circuit board is located between the heat source and the heat storage component. When the heat source generates relatively high heat instantly, the working medium inside the heat storage component absorbs the heat conducted from the heat source to the circuit board by latent heat of absorption during the phase change, and thus has excellent instant heat dissipation performance.

(11) In order to make the above-mentioned features and advantages of the present invention more comprehensible, the following specific embodiments are described in detail in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic perspective view of an electronic device according to an embodiment of the disclosure.

(2) FIG. 2 is a schematic perspective view of the electronic device of FIG. 1 from another perspective.

(3) FIG. 3 is a schematic cross-sectional view of partial components of the electronic device of FIG. 2 taken along a section line I-I.

(4) FIG. 4 is a schematic perspective view of an electronic device according to another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(5) FIG. 1 is a schematic perspective view of an electronic device according to an embodiment of the disclosure. FIG. 2 is a schematic perspective view of the electronic device of FIG. 1 from another perspective. FIG. 3 is a schematic cross-sectional view of partial components of the electronic device of FIG. 2 taken along a section line I-I. With reference to FIG. 1 to FIG. 3, in this embodiment, an electronic device **100** may be a part of a notebook computer (such as a host responsible for logical operations and data access) or other portable electronic device, and includes a case **110**, a circuit board **120**, and a heat dissipation structure **200**. The circuit board **120** and the heat dissipation structure **200** are disposed in the case **110**, and the heat dissipation structure **200** is disposed on the circuit board **120**. Various electronic components are disposed on the circuit board **120**, with its central processing unit (CPU) and graphics processing unit (GPU) as main heat sources. The heat dissipation structure **200** is thermally coupled to the main heat sources to rapidly extract the heat generated by the main heat sources out of the case **110** or to avoid the temperature inside the case **110** from rising sharply.

(6) In detail, the electronic device **100** of this embodiment further includes a first heat source **130** and a second heat source **140**. The first heat source **130** and the second heat source **140** are disposed on a first side **122** of the circuit board **120** and are thermally coupled to the circuit board **120**, and the first heat source **130** and the second heat source **140** may be one of the CPU and the GPU, respectively. On the other hand, the heat dissipation structure **200** includes a conductive member **220**, a first heat pipe **230**, a first fan **240**, a first heat storage component **250**, a second heat pipe **260**, a second fan **270**, and a second heat storage component **280**. The conductive member **220** may be composed of copper, aluminum, or other high thermal conductivity material, and is thermally coupled to the first heat source **130** and the second heat source **140** to extract out the heat generated by the first heat source **130** and the second heat source **140**, so as to provide a first heat dissipation path for the first heat source **130** and the second heat source **140**.

(7) It should be noted that the terms “first” and “second” mentioned in the disclosure are only used to distinguish different components and structures to facilitate intuitive understanding and reading, and are not intended to limit the disclosure.

(8) In detail, the conductive member **220** is disposed on the first heat source **130** and the second heat source **140**, and is located on the first side **122** of the circuit board **120**. The first heat source **130** is located between the conductive member **220** and the circuit board **120**, and the second heat source **140** is located between the conductive member **220** and the circuit board **120**.

(9) The first heat pipe **230** includes an evaporating part **232** and a condensing part **234**, and the second heat pipe **260** includes an evaporating part **262** and a condensing part **264**. The evaporation part **232** of the first heat pipe **230** and the evaporation part **262** of the second heat pipe **260** are disposed on the conductive member **220** and are thermally coupled to the conductive member **220**. The conductive member **220** is located between the evaporation part **232** of the first heat pipe **230** and the first heat source **130**, and the conductive member **220** is located between the evaporation part **262** of the second heat pipe **260** and the second heat source **140**. The first fan **240** is disposed in correspondence to an air outlet of the condensing part **234** of the first heat pipe **230**, and the second fan **270** is disposed in correspondence to an air outlet of the condensing part **264** of the second heat pipe **260**.

(10) Accordingly, the heat generated by the first heat source **130** may be conducted to the first heat pipe **230** through the conductive member **220**, and the heat in the first heat pipe **230** may be extracted out of the case **110** by heat exchange with an airflow generated by the first fan **240**. The heat generated by the second heat source **140** may be conducted to the second heat pipe **260** through the conductive member **220**, and the heat in the second heat pipe **260** may be extracted out of the case **110** by heat exchange with an airflow generated by the second fan **270**.

(11) Furthermore, the heat dissipation structure **200** further includes multiple first heat dissipation pastes **210**, which may employ liquid copper heat dissipation pastes, but the disclosure is not limited thereto. These first heat dissipation pastes **210** are disposed between the conductive member **220** and the first heat source **130** and between the conductive member **220** and the second heat source **140**. The conductive member **220** is thermally coupled to the first heat source **130** and the second heat source **140** through these first heat dissipation pastes **210**, and is fixed to the first heat source **130** and the second heat source **140** through these first heat dissipation pastes **210**.

(12) In this embodiment, the first heat storage component **250** and the second heat storage component **280** are disposed on a second side **124** of the circuit board **120** and are thermally coupled to the circuit board **120**. In other words, the circuit board **120** is located between the first heat source **130** and the first heat storage component **250**, and the circuit board **120** is located between the second heat source **140** and the second heat storage component **280**.

(13) The heat generated by the first heat source **130** is conducted from the first side **122** of the circuit board **120** to the conductive member **220**, then further conducted to the circuit board **120**, and finally conducted to the first heat storage component **250** through the second side **124** of the circuit board **120**. Similarly, the heat generated by the second heat source **140** is conducted from the second side **124** of the circuit board **120** to the conductive member **220**, then further conducted to the circuit board **120**, and finally conducted to the second heat storage component **280** through the second side **124** of the circuit board **120**, so as to provide a second heat dissipation path for the first heat source **130** and the second heat source **140**.

(14) In detail, the first heat storage component **250** may be a hollow case body composed of copper, aluminum, or other high thermal conductivity material, and the first heat storage component **250** may be filled with a first working medium **252**. The second heat storage component **280** may be a hollow case body composed of copper, aluminum, or other high thermal conductivity material, and the second heat storage component **280** may be filled with a second working medium **282**. Among them, the first working medium **252** and the second working medium **282** may be phase change materials (PCM).

(15) When the first heat source **130** generates high-wattage transient power instantly, the first heat dissipation path of the heat dissipation structure **200** is unable to rapidly provide heat dissipation to the first heat source **130**, so the heat is conducted to the first heat storage component **250** through the second heat dissipation path. After the first working medium **252** absorbs enough heat to reach a phase change temperature, the first working medium **252** undergoes a phase change and instantly absorbs the heat conducted from the first heat source **130** to the circuit board **120** by latent heat of absorption during the phase change. Therefore, the heat dissipation structure **200** of the disclosure has excellent instant heat dissipation performance.

(16) Similarly, when the second heat source **140** generates high-wattage transient power instantly, the second heat dissipation path of the heat dissipation structure **200** is unable to rapidly provide heat dissipation to the second heat source **140**, so the heat is conducted to the second heat storage component **280** through the second heat dissipation path. After the second working medium **282** absorbs enough heat to reach a phase change temperature, the second working medium **282** undergoes a phase change and instantly absorbs the heat conducted from the second heat source **140** to the circuit board **120** by latent heat of absorption during the phase change. In other words, the first working medium **252** and the second working medium **282** may store the latent heat by phase change and absorb the heat conducted from the first heat source **130** and the second heat source **140** to the circuit board **120** when storing the latent heat. Therefore, the heat dissipation structure **200** of the disclosure has excellent instant heat dissipation performance.

(17) On the other hand, after the electronic device **100** is shut down, the first working medium **252** and the second working medium **282** release the latent heat by phase change. Since the electronic device **100** has been shut down at this time, the working performance of the electronic device **100** is not affected.

(18) In this embodiment, the first working medium **252** and the second working medium **282** may employ 3M™ Fluorinert FC-72 electronic engineering liquid, which has a boiling point of 56° C. and requires to absorb 88 kJ/kg during phase change. In other words, the 3M™ Fluorinert FC-72 electronic engineering liquid may instantly absorb the heat conducted from the first heat source **130** and the second heat source **140** to the circuit board **120** by latent heat of evaporation during boiling (liquid-gas phase change). Since the boiling points of the first working medium **252** inside the first heat storage component **250** and the second working medium **282** inside the second heat storage component **280** are extremely low, the first working medium **252** and the second working medium **282** may instantly absorb a huge amount of heat and evaporate when reaching the boiling points to change from a liquid medium to a gaseous medium. This may solve the problem of relatively high heat instantly generated by the existing CPU or GPU due to an instant increase of transient power, avoiding the temperature inside the electronic device **100** from rising sharply.

(19) When the heat generated by the first heat source **130** and/or the second heat source **140** decreases or the temperature inside the electronic device **100** decreases, the temperature inside the first heat storage component **250** and/or the second heat storage component **280** may decrease below the boiling points of the first working medium **252** and/or the second working medium **282**. At this time, the gaseous medium returns to the liquid medium by condensation, and the released heat may be extracted out of the electronic device **100** through the first heat dissipation path.

(20) In addition, the electronic engineering liquid has insulating properties. When the first heat storage component **250** or the second heat storage component **280** are accidentally broken and cause a leakage of the electronic engineering liquid, since the electronic engineering liquid is not conductive, there is no risk of short-circuiting and damaging the electronic components. Furthermore, the low boiling point of the electronic engineering liquid allows it to rapidly evaporate without leaving residues when an accidental leakage occurs.

(21) In other embodiments, the working medium may employ 3M™ Fluorinert FC-87 electronic engineering liquid, which has a boiling point of 30° C. and requires to absorb 88 kJ/kg during phase change, 3M™ Fluorinert FC-84 electronic engineering liquid, which has a boiling point of

80° C. and requires to absorb 80 kJ/kg during phase change, 3M™ Fluorinert FC-77 electronic engineering liquid, which has a boiling point of 97° C. and requires to absorb 84 kJ/kg during phase change, or other phase change material such as a refrigerant.

(22) It should be noted that the first heat source **130** and the second heat source **140** of this embodiment may be set at different working temperatures, and the first working medium **252** and the second working medium **282** may employ different phase change materials to respectively maintain the first heat source **130** and the second heat source **140** at the working temperatures. In other words, the temperatures at which the first working medium **252** and the second working medium **282** undergo a phase change are different from each other. That is, this embodiment employs various electronic engineering liquids having different boiling points to optimize heat dissipation efficiency in accordance with different temperatures generated by various heat sources.

(23) In other embodiments, the first heat source and the second heat source are set at the same working temperature, and the first working medium and the second working medium may employ the same phase change material to maintain the first heat source and the second heat source at the same working temperature.

(24) In other embodiments, the working medium may employ different phase change materials according to requirements and store latent heat by phase change absorbing the heat generated by the heat source other than the liquid-gas phase change generating heat of vaporization, such as solid-liquid phase change generating heat of fusion, solid-gas phase change generating heat of sublimation, or other phase change between two different solid phases, and the disclosure does not limit the method.

(25) In this embodiment, the heat dissipation structure **200** further includes a second heat dissipation paste **294** and a third heat dissipation paste **296**. The second heat dissipation paste **294** and the third heat dissipation paste **296** may employ liquid copper heat dissipation pastes, but the disclosure is not limited thereto. The second heat dissipation paste **294** is disposed between the circuit board **120** and the first heat storage component **250**, and the third heat dissipation paste **296** is disposed between the circuit board **120** and the second heat storage component **280**. The circuit board **120** is thermally coupled to the first heat storage component **250** and the second heat storage component **280** through the second heat dissipation paste **294** and the third heat dissipation paste **296**, respectively.

(26) In addition, the first heat storage component **250** and the second heat storage component **280** include multiple fastening holes **292**. The heat dissipation structure **200** further includes multiple fastening members **298**. The fastening member **298** is a screw, and the screw sequentially passes through the conductive member **220** and the circuit board **120** to be fastened to the fastening hole **292** of the first heat storage component **250** and the second heat storage component **280**. In other words, in addition to providing the second heat dissipation path for the first heat source **130** and the second heat source **140**, the first heat storage component **250** and the second heat storage component **280** may also be used for fastening purposes.

(27) To put it another way, the disclosure replaces the traditional metal backplate with the first heat storage component **250** and the second heat storage component **280**. In addition, the first heat storage component **250** and the second heat storage component **280** are thermally coupled to the second side **124** of the circuit board **120** so that the heat generated by the first heat source **130** and the second heat source **140** may be dissipated from the second side **124** of the circuit board **120** through the second heat dissipation path. Moreover, the first heat storage component **250** and the second heat storage component **280** have the function of the traditional metal backplate, which may avoid solder cracks or die cracks caused by excessive bending of the circuit board **120** when the first heat source **130** and the second heat source **140** are fastened to the conductive member **220** or other heat dissipation module. The first heat storage component **250** and the second heat storage component **280** may also enhance strength and balance of fastening the conductive member **220** or other heat dissipation module to reduce contact thermal resistance.

(28) FIG. 4 is a schematic perspective view of an electronic device according to another embodiment of the disclosure. With reference to FIG. 4, the difference between an electronic device **100A** of this embodiment and the electronic device **100** of the previous embodiment lies in that a heat dissipation structure **200A** of this embodiment provides a single heat storage component **240A**, which may correspond to multiple heat sources. For example, the heat storage component **240A** corresponds to the first heat source and the second heat source of the previous embodiment at the same time. The heat storage component **240A** has a relatively huge volume and may be filled with more working medium to meet higher power consumption and heating requirements.

(29) In summary, in the heat dissipation structure and the electronic device of the disclosure, the circuit board is thermally coupled to the heat source, the heat storage component is further thermally coupled to the circuit board, and the circuit board is located between the heat source and the heat storage component. When the heat source generates relatively high heat instantly, the working medium inside the heat storage component absorbs the heat conducted from the heat source to the circuit board by latent heat of absorption during the phase change, and thus has excellent instant heat dissipation performance.

(30) Although the disclosure has been described with reference to the above embodiments, they are not intended to limit the disclosure. It will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit and the scope of the disclosure. Accordingly, the scope of the disclosure will be defined by the attached claims and their equivalents and not by the above detailed descriptions.

Claims

1. A heat dissipation structure, comprising: a conductive member, disposed on at least one heat source and thermally coupled to the at least one heat source; at least one heat pipe, comprising an evaporation part and a condensing part, wherein the evaporation part of the at least one heat pipe is disposed on the conductive member and is thermally coupled to the conductive member; at least one fan, corresponding to the condensing part of the at least one heat pipe; at least one heat storage component, disposed on a circuit board, wherein the at least one heat source is located between the conductive member and the circuit board, the conductive member is located between the at least one heat pipe and the at least one heat source, and the circuit board is located between the at least one heat source and the at least one heat storage component, wherein the circuit board is thermally coupled to the at least one heat source, the at least one heat storage component is thermally coupled to the circuit board, the at least one heat storage component is filled with a working medium, and the working medium absorbs heat conducted from the at least one heat source to the circuit board by latent heat of absorption during phase change, the circuit board has a first side and a second side opposite to the first side, the at least one heat source via the conductive member, the at least one heat pipe and the at least one fan located on the first side to form a first heat dissipation path, the at least one heat source via the circuit board and the at least one heat storage component located on the second side to form a second heat dissipation path, and the heat generated by the at least one heat source is dissipated from the first heat dissipation path and the second heat dissipation path located on opposite two sides of the at least one heat source, under a first power, the heat is conducted to the at least one heat storage component through the second heat dissipation path, and the working medium filled in the at least one heat storage component undergoes phase change to absorb and store the heat conducted through the second heat dissipation path, until a temperature inside the at least one heat storage component decreases below a boiling point of the working medium, the heat stored in the working medium is unable to release, as the first power is decreased to a second power lower than the first power, the working medium undergoes phase change to release heat stored therein due to the temperature inside the at least one heat storage component decreasing below the boiling point of the working medium, and the heat released from the working

medium is conducted to the first heat dissipation path.

2. The heat dissipation structure according to claim 1, wherein the number of the at least one heat source is a plurality, the number of the at least one heat storage component is a plurality in correspondence to the number of the heat sources, and each of the heat storage components is thermally coupled to a corresponding one of the heat sources.

3. The heat dissipation structure according to claim 2, temperatures at which the working media inside the heat storage components undergo phase change are different from each other.

4. The heat dissipation structure according to claim 1, wherein the number of the at least one heat source is a plurality, the number of the at least one heat storage component is one, and the heat storage component is thermally coupled to the heat sources.

5. The heat dissipation structure according to claim 1, wherein the at least one heat storage component comprises a plurality of fastening holes, the heat dissipation structure further comprises a plurality of fastening members, and the fastening members pass through the conductive member and the circuit board to be fastened to the fastening holes.

6. An electronic device, comprising: a case; a circuit board, disposed in the case; at least one heat source, disposed on the circuit board; and a heat dissipation structure, comprising: a conductive member, disposed on the at least one heat source and thermally coupled to the at least one heat source, wherein the at least one heat source is located between the conductive member and the circuit board; at least one heat pipe, comprising an evaporating part and a condensing part, wherein the evaporating part of the at least one heat pipe is disposed on the conductive member and is thermally coupled to the conductive member, and the conductive member is located between the at least one heat pipe and the at least one heat source; at least one fan, disposed in correspondence to the condensing part of the at least one heat pipe; and at least one heat storage component, disposed on the circuit board and thermally coupled to the circuit board, wherein the circuit board is located between the at least one heat source and the at least one heat storage component, the circuit board is thermally coupled to the at least one heat source, the at least one heat storage component is thermally coupled to the circuit board, the at least one heat storage component is filled with a working medium, and the working medium absorbs heat conducted from the at least one heat source to the circuit board by latent heat of absorption during phase change, the circuit board has a first side and a second side opposite to the first side, the at least one heat source via the conductive member, the at least one heat pipe and the at least one fan located on the first side to form a first heat dissipation path, the at least one heat source via the circuit board and the at least one heat storage component located on the second side to form a second heat dissipation path, and the heat generated by the at least one heat source is dissipated from the first heat dissipation path and the second heat dissipation path located on opposite two sides of the at least one heat source, under a first power, the heat is conducted to the at least one heat storage component through the second heat dissipation path, and the working medium filled in the at least one heat storage component undergoes phase change to absorb and store the heat conducted through the second heat dissipation path, until a temperature inside the at least one heat storage component decreases below a boiling point of the working medium, the heat stored in the working medium is unable to release, as the first power is decreased to a second power lower than the first power, the working medium undergoes phase change to release heat stored therein due to the temperature inside the at least one heat storage component decreasing below the boiling point of the working medium, and the heat released from the working medium is conducted to the first heat dissipation path.

7. The electronic device according to claim 6, wherein the number of the at least one heat source is a plurality, the number of the at least one heat storage component is a plurality in correspondence to the number of the heat sources, and each of the heat storage components is thermally coupled to a corresponding one of the heat sources.

8. The electronic device according to claim 7, temperatures at which the working media inside the heat storage components undergo phase change are different from each other.

9. The electronic device according to claim 6, wherein the number of the at least one heat source is a plurality, the number of the at least one heat storage component is one, and the heat storage component is thermally coupled to the heat sources.

10. The electronic device according to claim 6, wherein the at least one heat storage component comprises a plurality of fastening holes, the heat dissipation structure further comprises a plurality of fastening members, and the fastening members pass through the conductive member and the circuit board to be fastened to the fastening holes.
